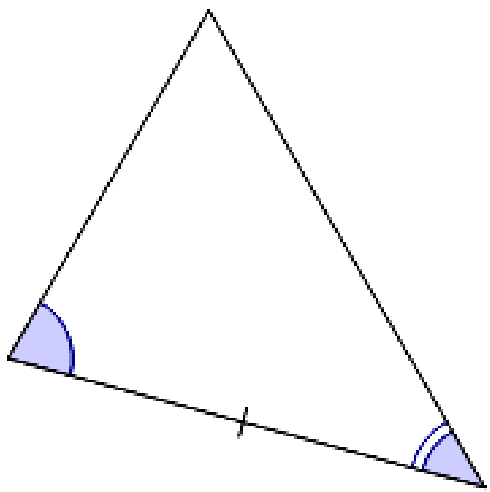
Review: Identifying congruent triangles

1 Given the following triangles, identify which two triangles are congruent.

A



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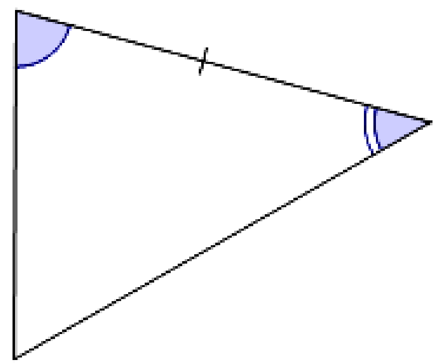

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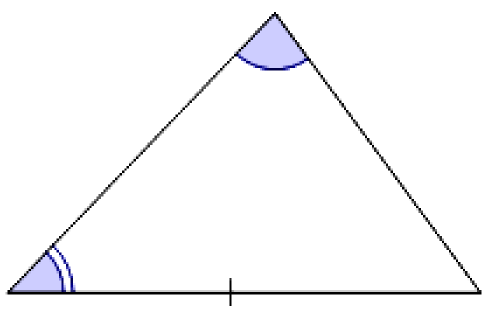
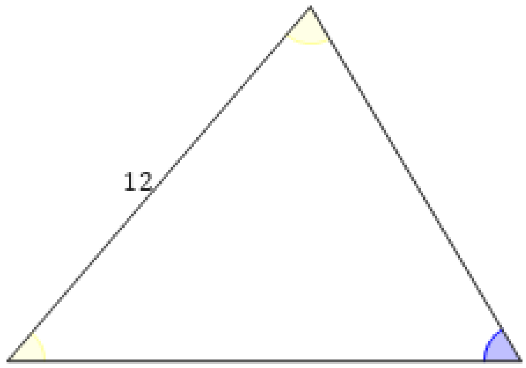


2 Given the following triangles, determine a p[redacted] congruent triangles.

A



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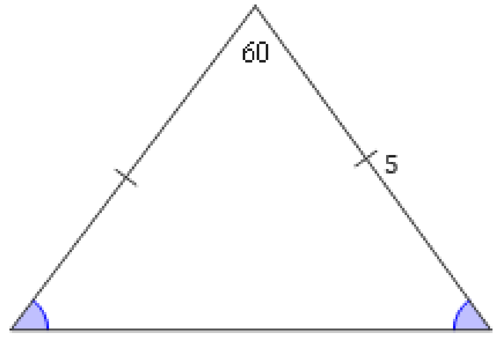

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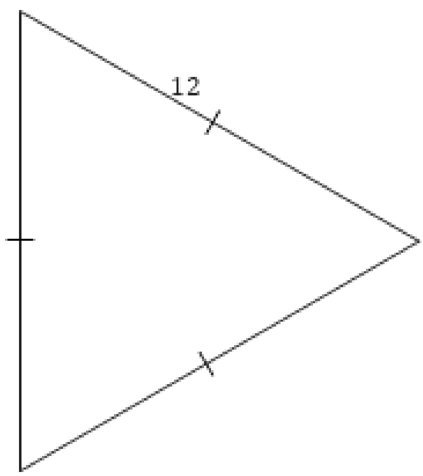

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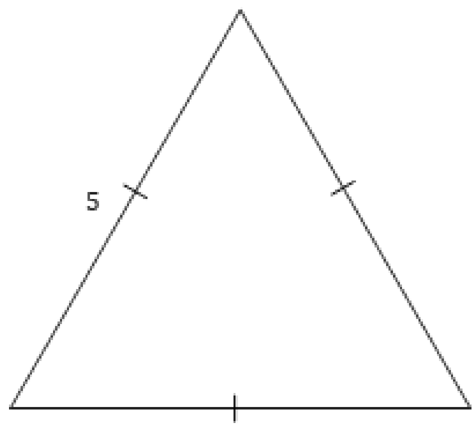


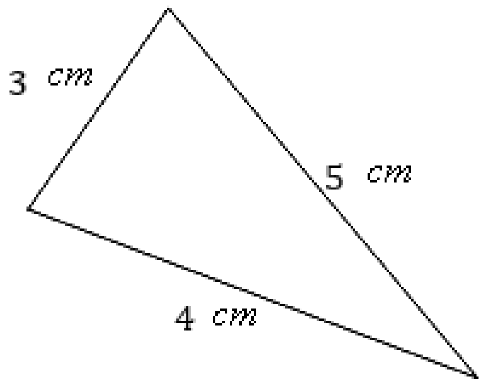
3 For each of following sets of three triangles

i Identify which triangles are congruent.

ii Identify the congruence theorem or postulate that justifies the congruence.

a

A



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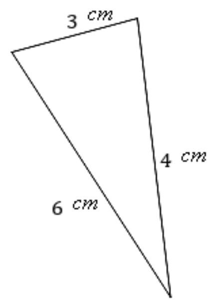

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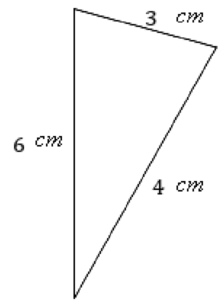
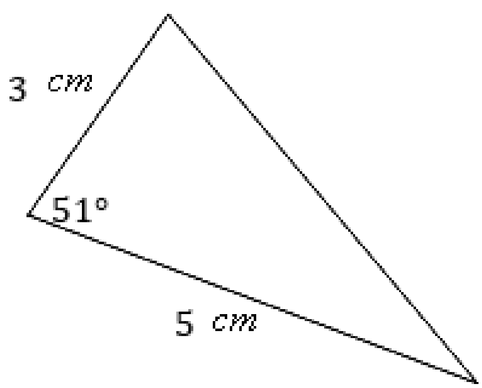


b

A



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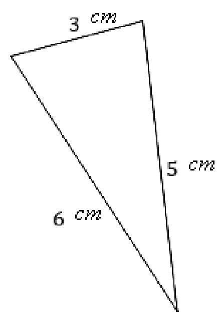

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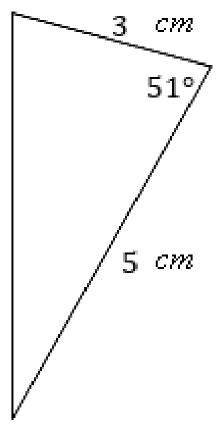
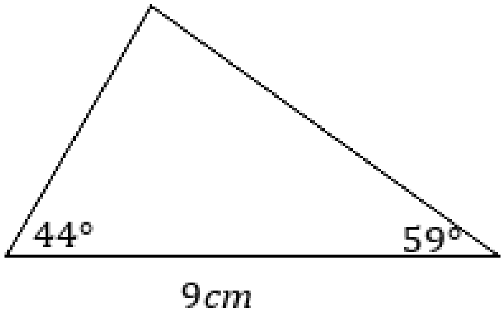


C

A



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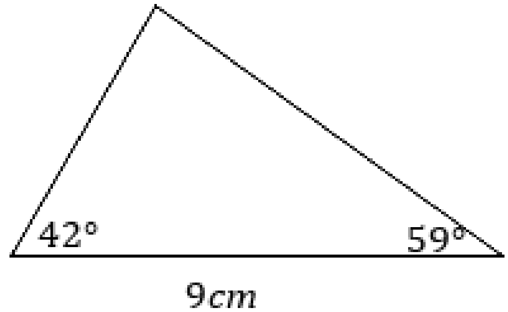

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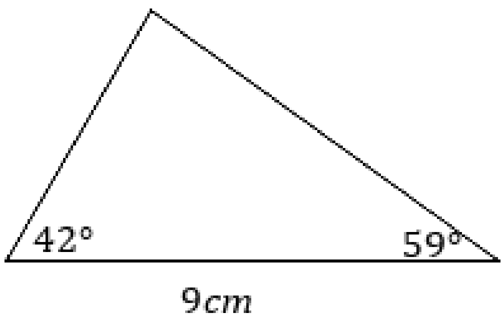
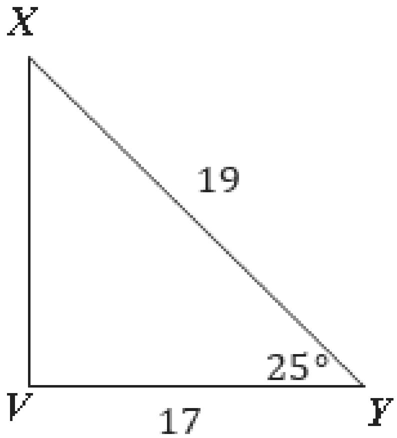


d

A



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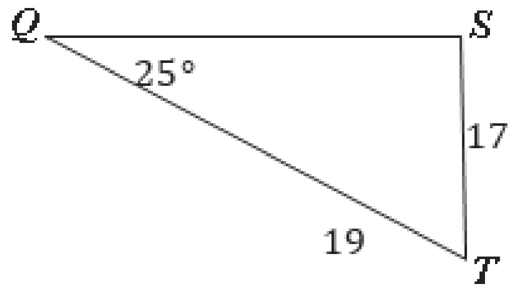

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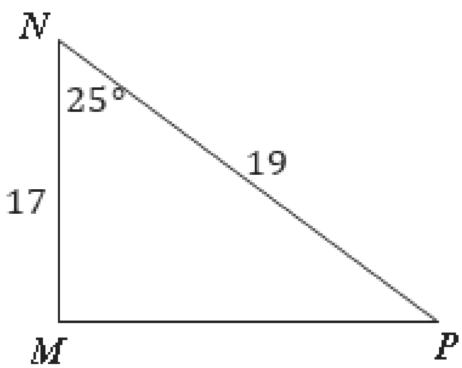


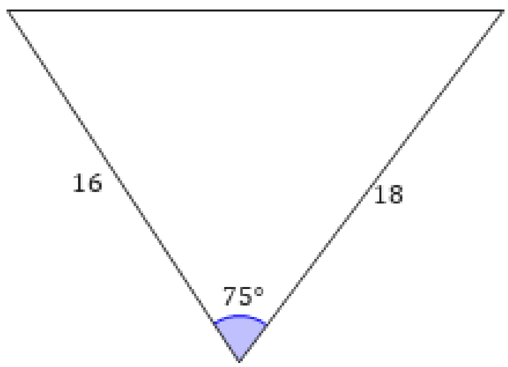
4 For each of the following sets of four triang

i Identify which pair of triangles are congruent.

ii Identify the congruence theorem or postulate that justifies the congruence.

a

A



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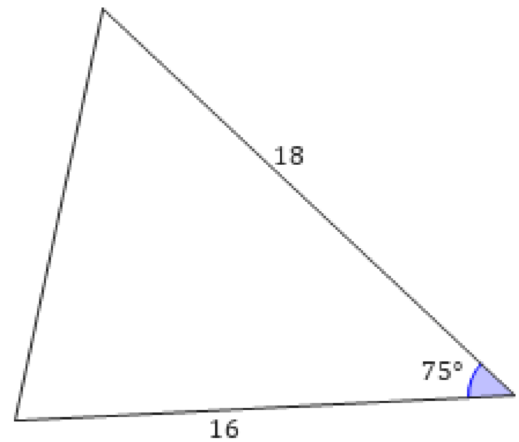

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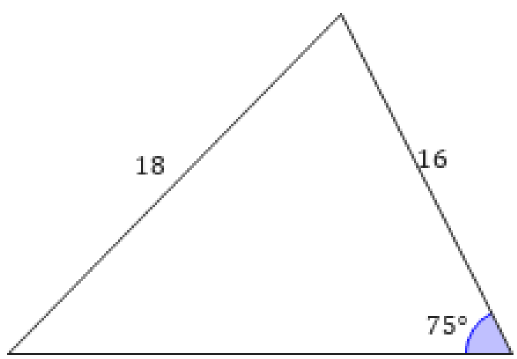

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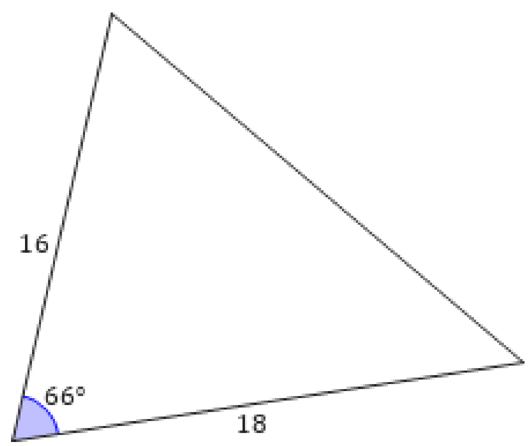
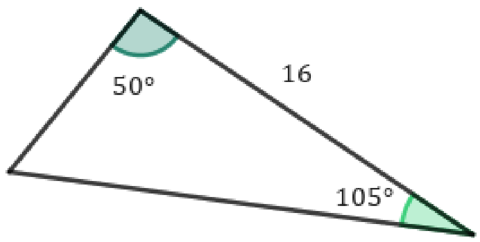


b

A



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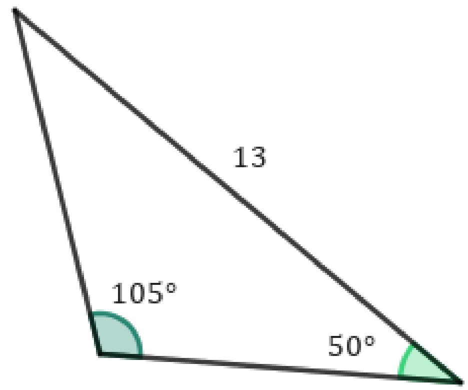

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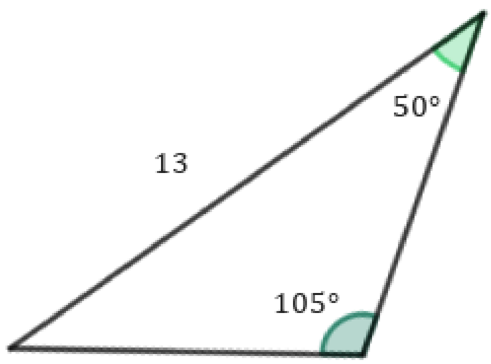

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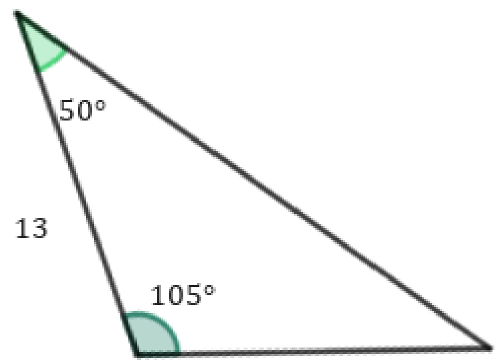

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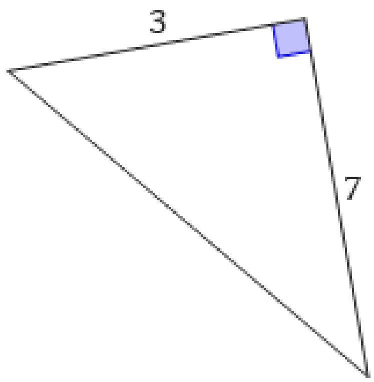
A



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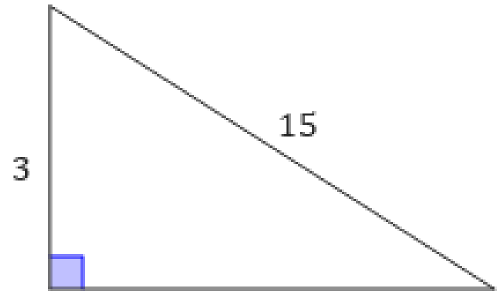

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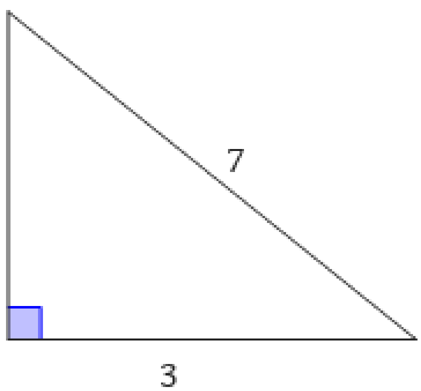

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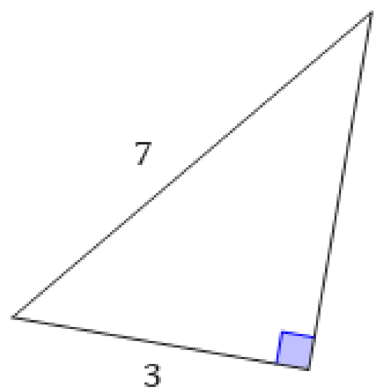
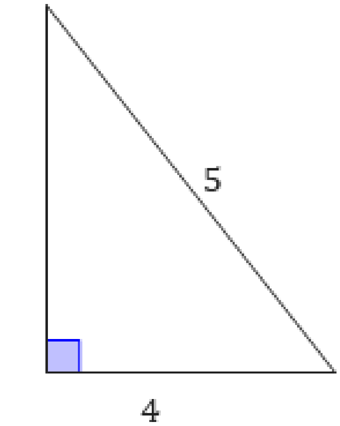


d

A



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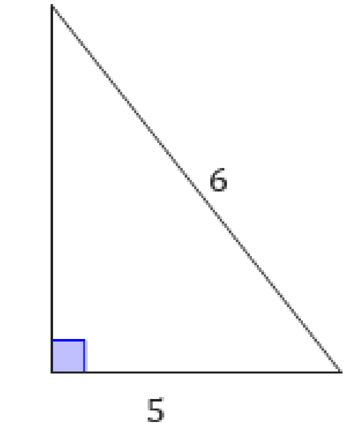

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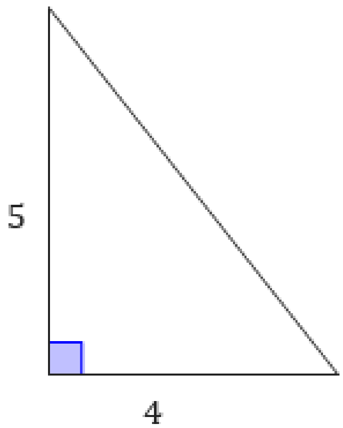

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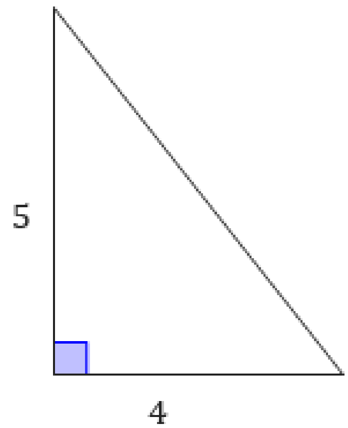
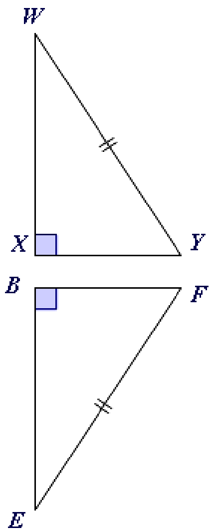


5 For each of the following, state whether the[redacted]enough information to prove that the pair of triangles are congruent.

a



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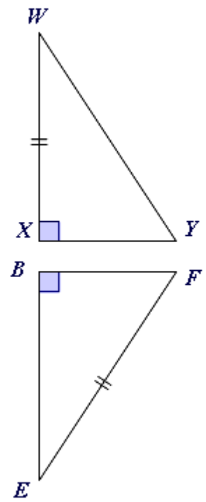

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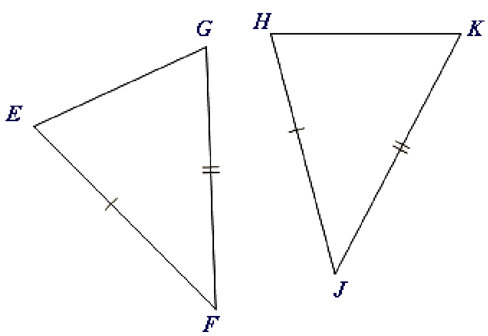

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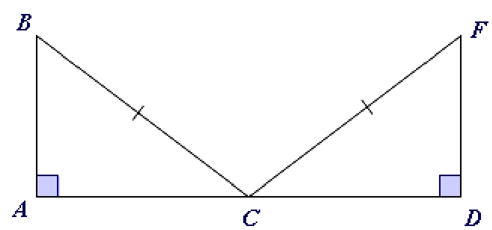
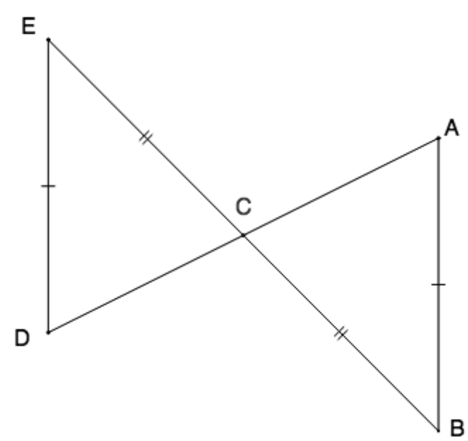
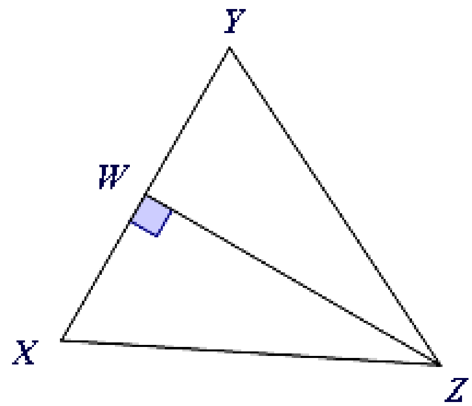


6 Consider the given triangles.
Do we have enough information to deduce that $\triangle CDE$ is congruent to $\triangle CAB$?



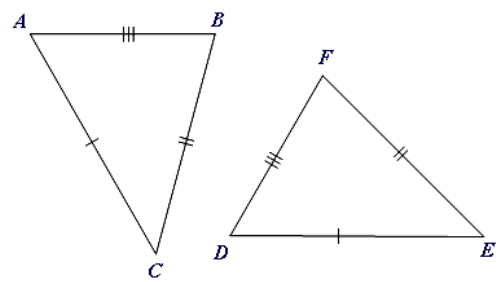
7 Consider the adjacent diagram.
Is there enough information to prove that $\triangle ZWY$ is congruent to $\triangle ZWX$?



8 Consider the adjacent diagram.

a From the information given on the diagram, which side is congruent to $\overline{AB}$?

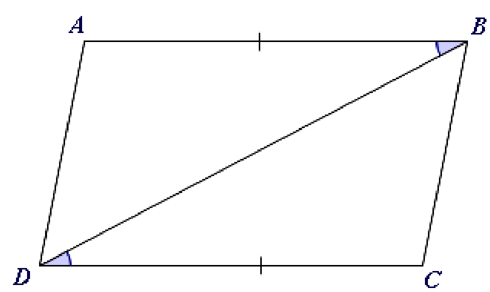
b State the congruence theorem or postulate that explains why $\triangle ABC$ is congruent to $\triangle DFE$.



9 Consider the adjacent diagram.

a From the information given on the diagram, which angle is congruent to $\angle ABD$?

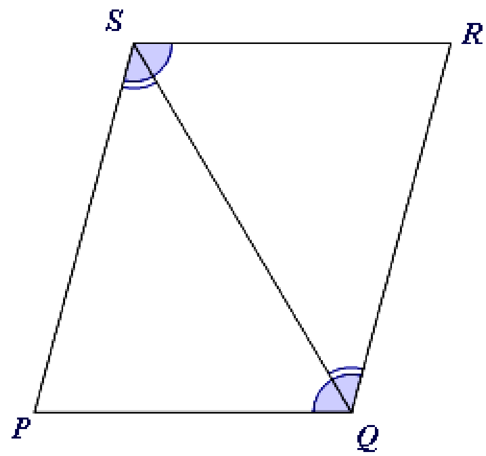
b State the congruence theorem or postulate that explains why $\triangle ADB$ is congruent to $\triangle CBD$.



10 Consider the adjacent diagram.

a From the information given on the diagram, which angle is congruent to $\angle PSQ$?

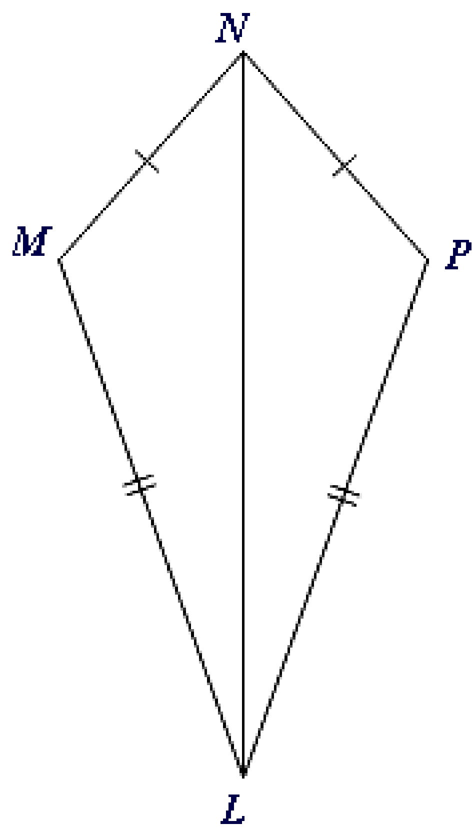
b Identify the congruence theorem or postulate that explains why $\triangle PSQ$ is congruent to $\triangle RQS$.



11 Consider the adjacent diagram.

a From the information given on the diagram, which side is congruent to $\overline{MN}$?

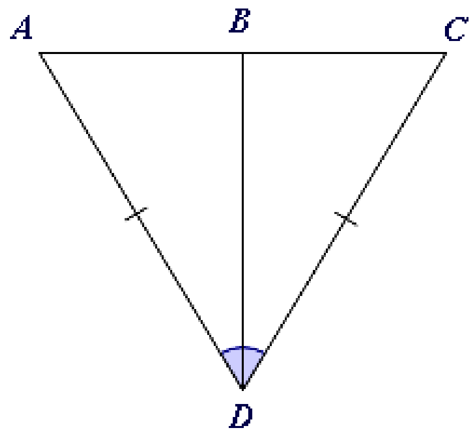
b Identify the congruence theorem or postulate that explains why $\triangle LNM$ is congruent to $\triangle LNP$.



12 Consider the adjacent diagram:

a From the information given on the diagram, which angle is congruent to $\angle ADB$?

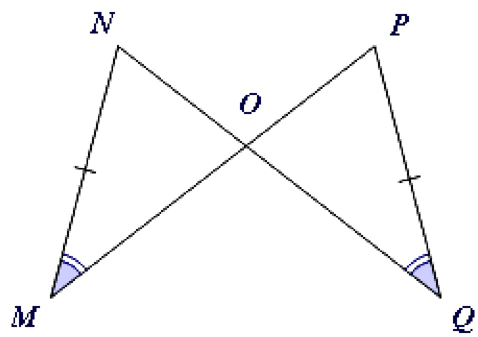
b Identify the congruence theorem or postulate that explains why $\triangle ABD$ is congruent to $\triangle CBD$.



13 Consider the adjacent diagram.

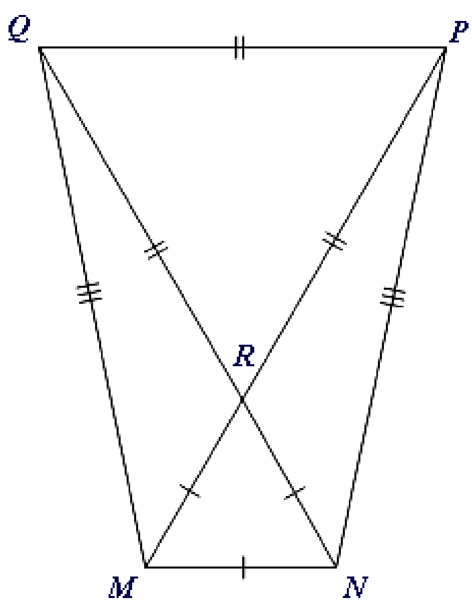
a Identify which angle is congruent to $\angle NOM$.

b Identify the congruence theorem or postulate that explains why $\triangle NOM$ is congruent to $\triangle POQ$.

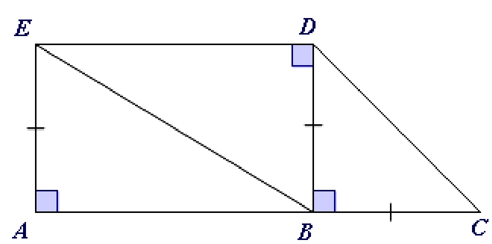


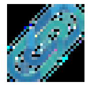
14 For each of the following diagrams, identify two triangles that are congruent.

a

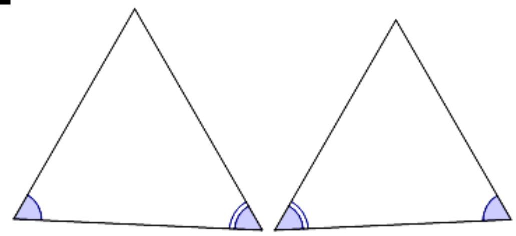


b





15 Consider the following diagram.
Explain why the information given is not enough to deduce that the triangles are congruent.

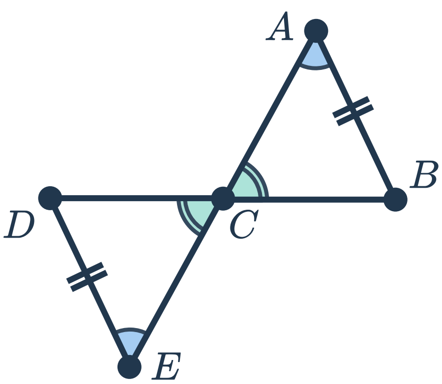


16 Consider two triangles that have two pairs of congruent sides and one pair of congruent angles.
Is this enough to prove that the triangles are congruent?

Additional questions

17 Determine which of the following pairs of shapes are congruent. For each congruent pair, state the transformation or series of transformations that maps one shape to the other.

a



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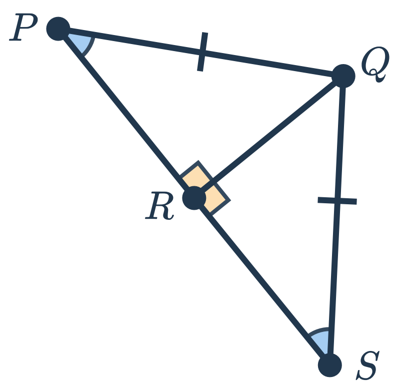
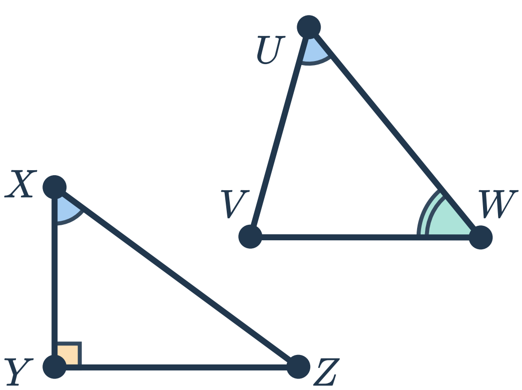
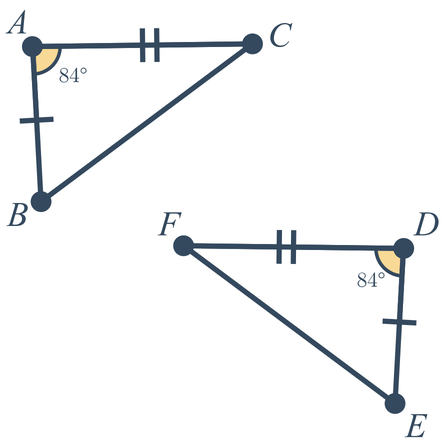


c



18 For each of the following pairs of triangles, [redacted]fy a congruence theorem or postulate that justifies the congruence. If a theorem or postulate does not exist, determine whether the triangles are not congruent or if there is not enough information to justify the congruence.

a



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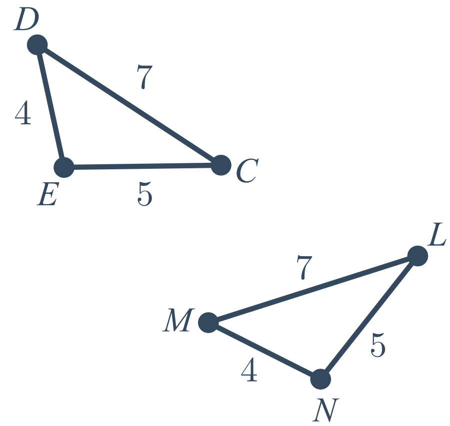

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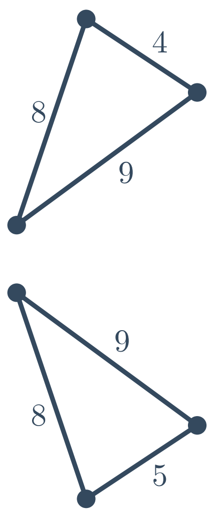

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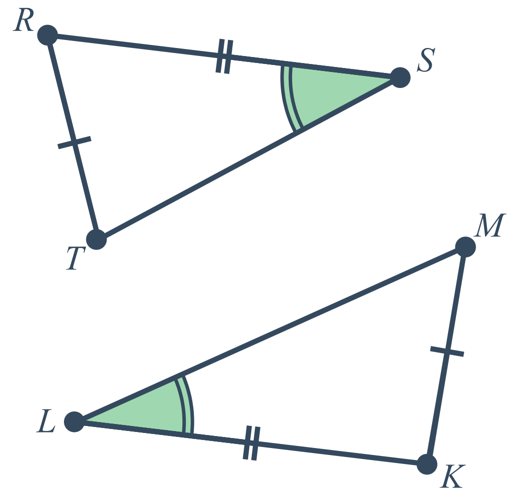
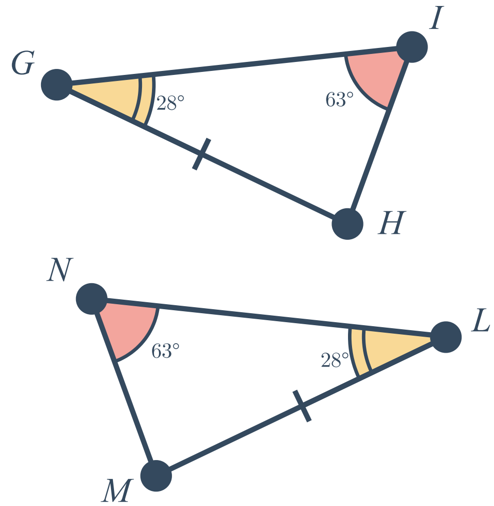


e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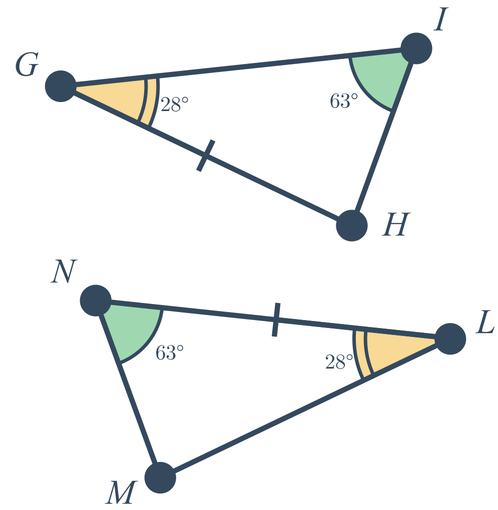
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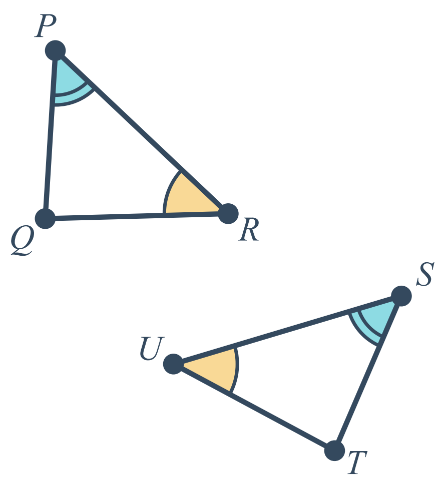
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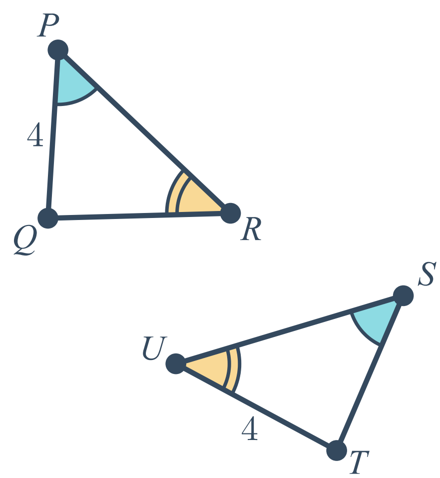
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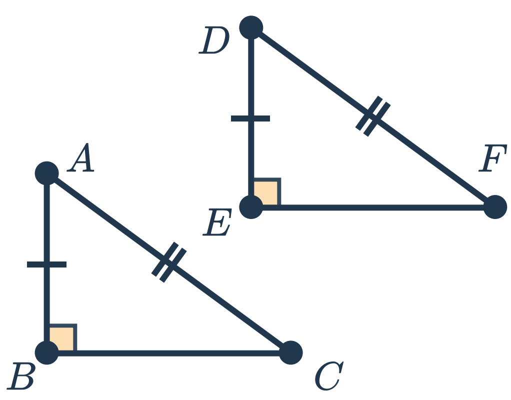
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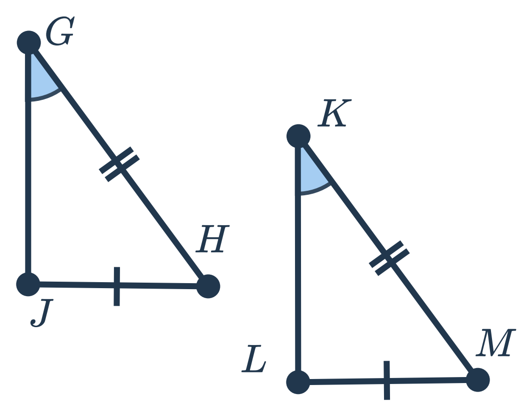
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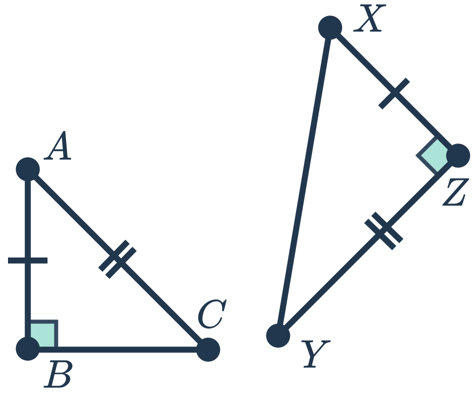
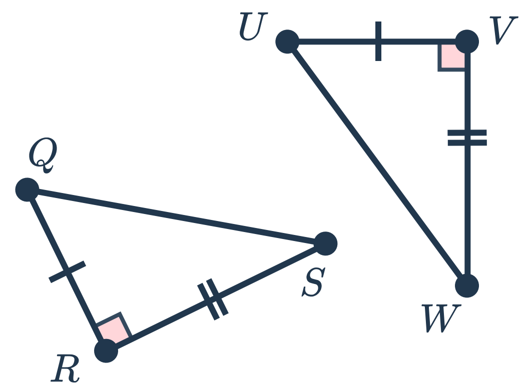
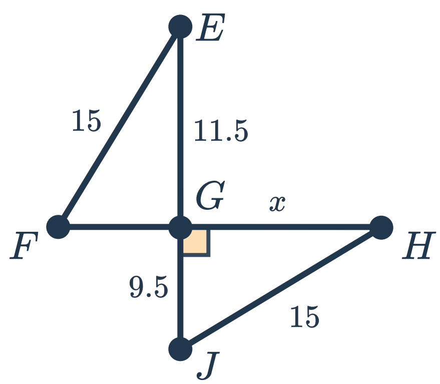




19 Find the value of the variable(s) that makes the triangles congruent.

a



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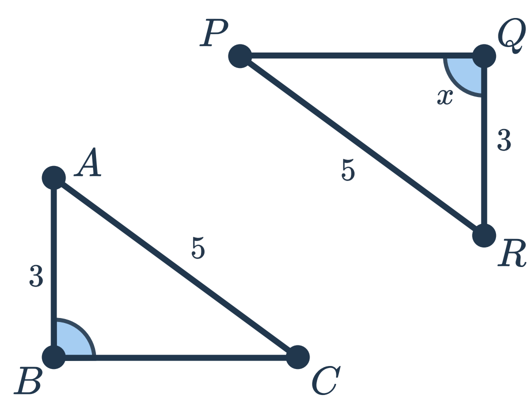
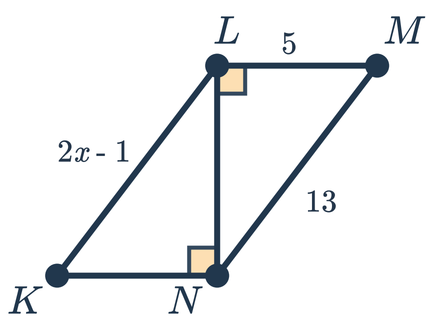
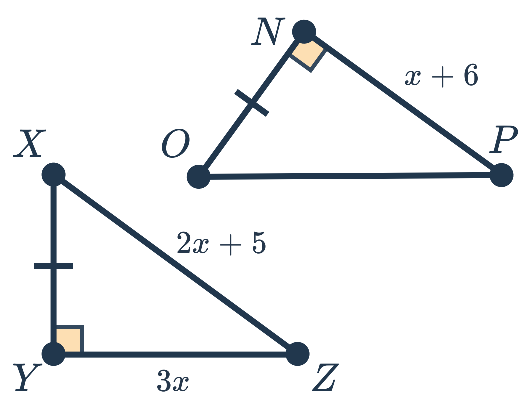


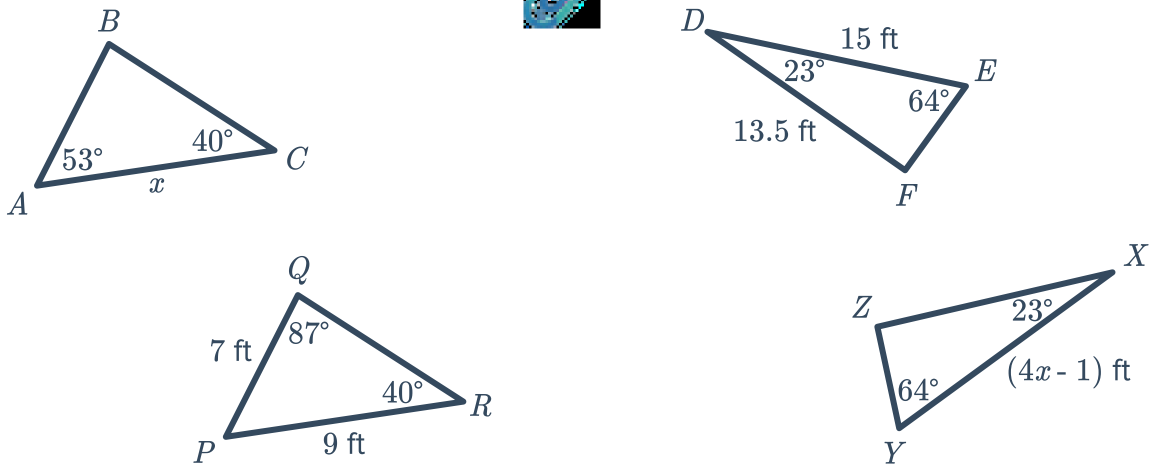
c



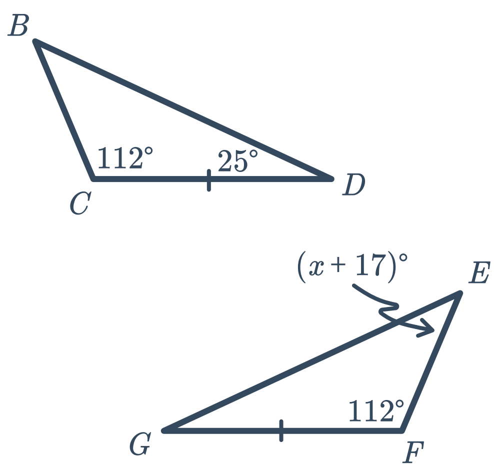
d



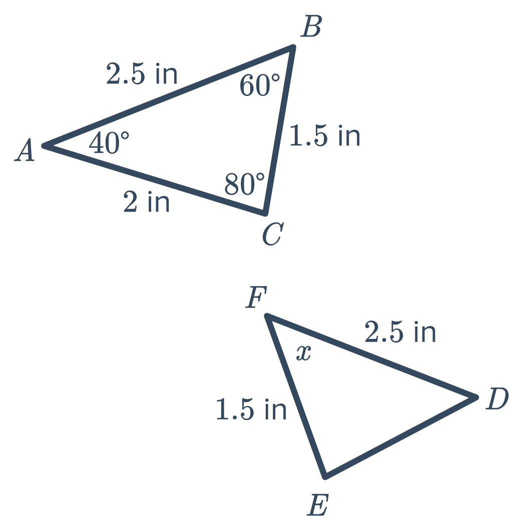
e


B
A
C
53°
40°
x
D
E
F
15 ft
23°
64°
13.5 ft
Q
P
R
87°
7 ft
40°
9 ft
Z
X
Y
23°
64°
(4x - 1) ft

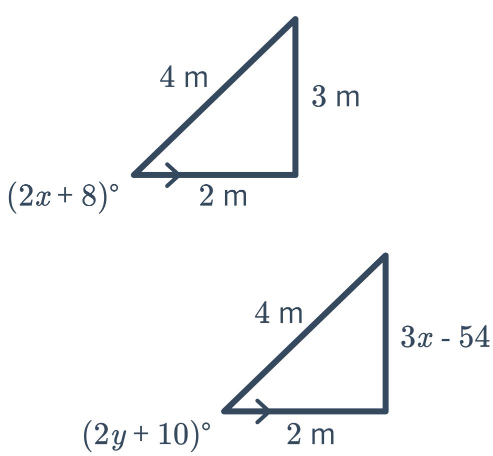
g


B
C
D
112°
25°
(x + 17)°
E
G
F
112°

h

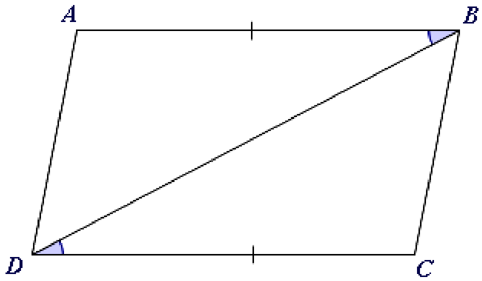

B
A
C
2.5 in
60°
1.5 in
40°
80°
2 in
F
D
E
2.5 in
x
1.5 in

i

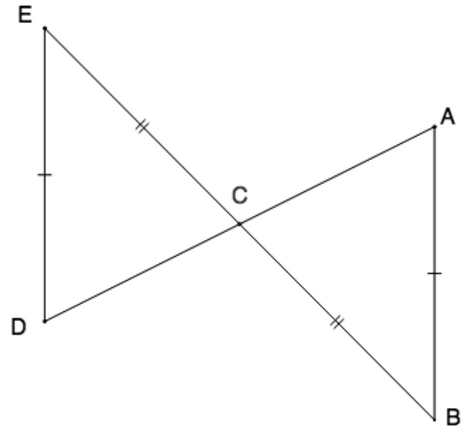

4 m
3 m
(2x + 8)°
2 m
4 m
3x - 54
(2y + 10)°
2 m

20 Is there enough information to determine w[redacted]r the triangles are congruent? If not, list a pair of congruent parts that would guarantee triangle congruence and name the postulate that justifies it.

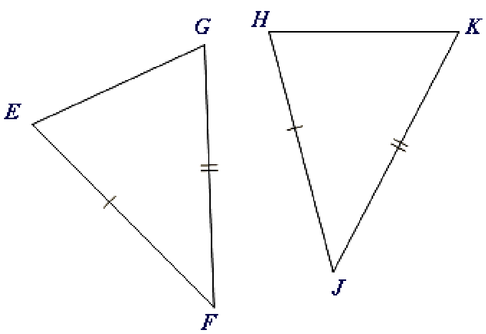
a



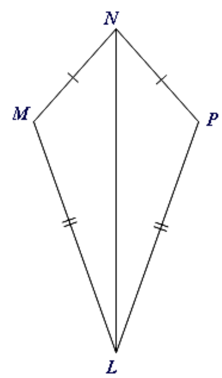
b



c



d

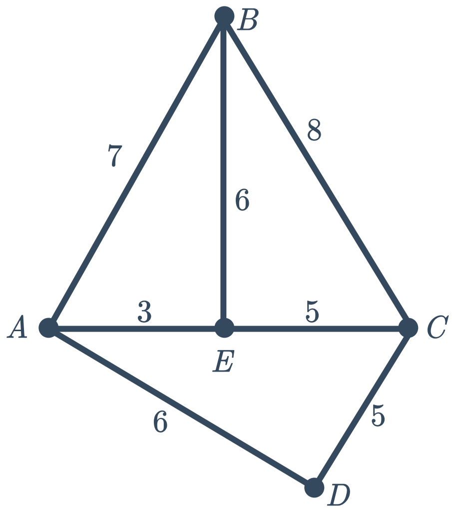


21 Use the given diagram to

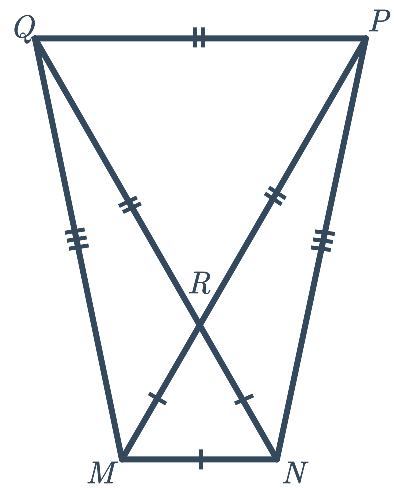
i State which two triangles are congruent

ii State the congruence postulate the triangles satisfy

a



b



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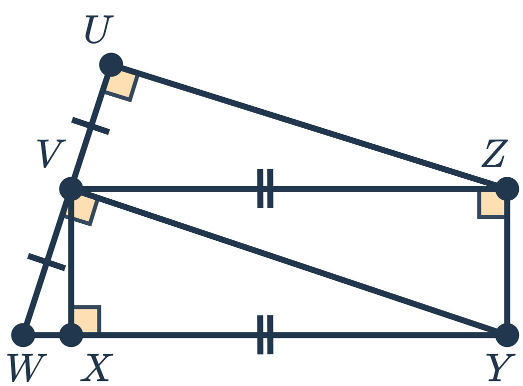
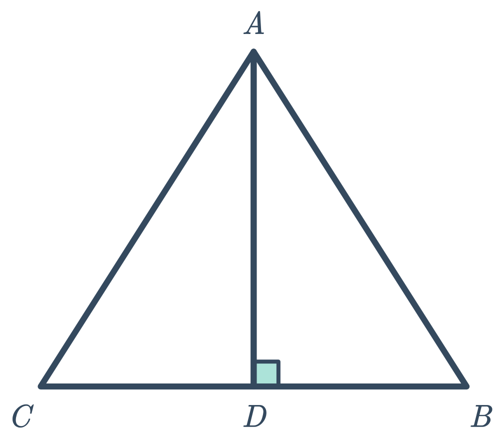


22 Identify a pair of additional congruent parts [redacted]ed to prove the triangles congruent by

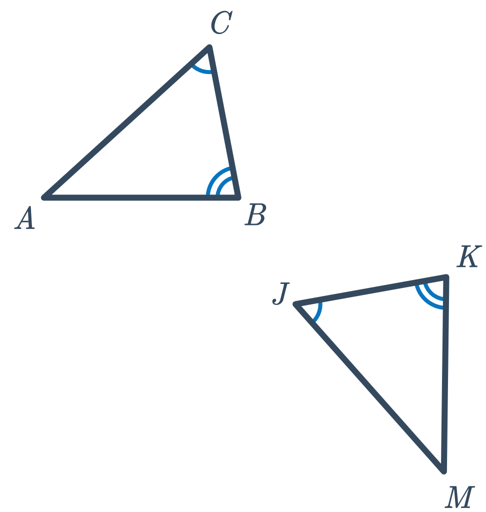
i Angle-angle-side congruence (AAS)

ii Angle-side-angle congruence (ASA)

a $\triangle ADC \cong \triangle ADB$

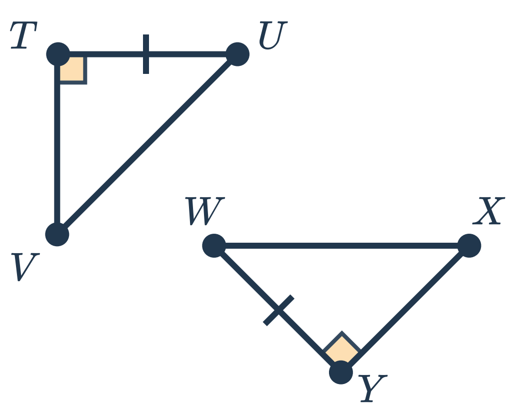


b $\triangle ACD \cong \triangle CAB$

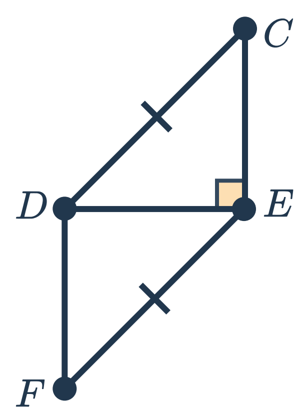


23 Identify a pair of additional congruent parts needed to prove the triangle pairs are congruent by the Hypotenuse-leg (HL) congruency theorem.

a $\triangle TUV \cong \triangle WXY$



b $\triangle CDE \cong \triangle FED$



24 If two line segments bisect each other, and their endpoints are joined, will a pair of congruent triangles always be formed? Explain.

25 Fleur and Jeung have each drawn a triangle. The longest sides of their triangles have the same length, as do the shortest sides of their triangles.
They claim that their triangles must be congruent by the Hypotenuse-leg (HL) congruency theorem, since the longest side of a triangle is always the hypotenuse. Are they correct?

26 A teacher provides his class of Geometry students with two angles and one side of an unknown triangle. Will the students have enough information to construct a congruent triangle? Explain.